

Damped Correlated Matching: consolidated mechanisms, laws, and proofs

Rotated surface code, Z-memory, circuit-level depolarizing noise; PyMatching fork with enable_correlations and Pearl soft-evidence α . All results reproduced by the scripts and notebooks listed in the appendix.

Abstract. Full-strength correlated matching (CM, Fowler 2013) *loses one unit of fault-tolerant distance*: at $d = 5$ it fails on two-fault configurations that MWPM corrects, acquiring a logical-error floor $\propto p^2$ that makes it cross *above* plain MWPM at low physical error rate. The failing configurations are, exhaustively, two-qubit-gate (CNOT) correlated faults whose first-pass selection donates a cloud of near-free “phantom” edges through which the second pass completes an observable-crossing detour. Damping the reweighting with a Pearl soft-evidence factor α re-prices the phantom edges and restores the distance; the optimal α follows a saturating law $\alpha^*(p) = L / (1 + (p_0/p)^m)$ whose three parameters map onto mechanistic quantities. Every claim below is tied to an exhaustive fault enumeration, a Monte-Carlo sweep, or a control experiment.

1. Setup and the damped reweighting rule

MWPM decodes the X- and Z-detector graphs independently with prior edge weights $w_e = \log((1-p_e)/p_e)$. CM performs two passes: after a first MWPM pass selects edges, every edge μ that shares a DEM hyperedge mechanism (data-qubit Y, or mixed two-qubit CNOT error) with a selected edge ν is reweighted using the conditional probability. Pearl's soft-evidence rule generalises this with a confidence α :

$$p_\mu^{\text{imp}} = \alpha P(e_\mu | e_\nu) + (1 - \alpha) P(e_\mu | \neg e_\nu), \quad w_\mu^{\text{imp}} = \log \frac{1 - p_\mu^{\text{imp}}}{p_\mu^{\text{imp}}},$$

with $P(e_\mu | e_\nu) = \text{corr}(\mu, \nu) / \text{occ}(\nu)$ from DEM co-occurrence statistics; the strongest boost over selected neighbours is kept, and the prior is kept whenever the posterior is smaller. $\alpha = 1$ is ordinary (hard-evidence) CM; $\alpha \rightarrow 0$ recovers MWPM. Two exact reductions were verified: (i) at low p the second Pearl term is $O((1-\alpha)p)$ and decode-irrelevant — Pearl $\equiv$ multiplicative damping (0 decode disagreements at $p \leq 10^{-3}$); (ii) the conditional rule is Fowler's uniform-rescaling rule — identical joint, normaliser differs at $O(p)$ (sum Z_ν vs odd-parity marginal $\text{occ}(\nu)$), giving 662 near-tie flips per 10^6 shots at $p = 10^{-2}$ with *no* LER change (6.990 vs 6.992×10^{-2}), and $0/10^6$ at $p = 4 \times 10^{-4}$. PyMatching's built-in sides with the conditional (1 vs 55 disagreements per 10^5).

2. The empirical laws (Monte-Carlo sweeps, common random numbers)

Sweeps over (d, p, α) with 500-error stopping (10^7 – 3×10^9 shots per point) give three laws:

law	measurement
$\alpha^*(p)$ rises with p and saturates	$d=5$: $\alpha^* = 0.1$ ($p=10^{-4}$) $\rightarrow$ 0.7 – 0.8 ($p=10^{-2}$); saturating fit $L/(1+(p_0/p)^m)$: $L=0.775$, $p_0 \approx 6 \times 10^{-4}$ (adj- $R^2=0.95$); $d=7$: $L=0.749$, $p_0 \approx 7 \times 10^{-4}$ (adj- $R^2=0.94$). Same plateau at both distances.
hard CM crosses above MWPM at low p	$d=5$, $p=10^{-4}$: $\text{LER}(\text{CM}, \alpha=1) / \text{LER}(\text{MWPM}) = 1.224$. Local log-log slope of $\text{CM}(1)$ flattens to 2.51–2.65 while MWPM holds ≈ 3.0 ; the excess channel scales $\approx p^2$.
tuned CM is a constant-factor gain	best- α /MWPM = 0.55 ± 0.03 across a full decade ($d=5$) and 0.31 – 0.32 ($d=7$): same slope as MWPM, constant prefactor. Hard CM instead changes the <i>slope</i> .

3. Proof of effective-distance loss: exhaustive fault enumeration

Every DEM mechanism ($M = 1677$ at $d = 5$; 6023 at $d = 7$) and every pair of mechanisms is injected combinatorially — the syndrome is built by XOR, ground truth is exact, no sampling. All decoders correct every weight-1 configuration. For weight-2:

configuration set	MWPM	CM ($\alpha=1$)	Pearl (α^*)
$d=5, p=2 \times 10^{-4}$: 1 405 326 pairs — failing pairs	0	373	14 ($\alpha^*=0.2$)
...probability weight $\sum p_i p_j$ (the p^2 LER floor)	0	2.80×10^{-7}	7.1×10^{-9}
$d=5, p=4 \times 10^{-4}$: failing pairs / weight	0	$337 / 9.77 \times 10^{-7}$	$8 / 1.4 \times 10^{-8}$
$d=7, p=4 \times 10^{-4}$: 18 135 253 pairs — failing pairs	0	0	0 ($\alpha^*=0.3$)

MWPM preserves the full code distance (all weight- ≤ 2 chains corrected); **hard CM's fault distance at $d = 5$ is 2**, i.e. one unit below MWPM's $\lfloor d/2 \rfloor = 3$; at $d = 7$ the loss is $4 \rightarrow 3$ (no 2-fault failures; excess slope ≈ 3). Quantitative closure at $p = 10^{-4}$:

$$7.9 \times 10^{-8} (\text{best}\alpha, \text{ meas.}) + 7.5 \times 10^{-8} (W_2, \text{ enum.}) = 1.54 \times 10^{-7} \approx 1.68 \times 10^{-7} (\text{CM}(1), \text{ meas.})$$

— 92% of hard CM's low- p LER decomposes into “damped-CM part + enumerated 2-fault channel”. Solving $W_2(p) = (1 - 0.575) \cdot \text{LER}_{\text{MWPM}}(p)$ predicts the crossover at $p^* = 1.3 \times 10^{-4}$, matching the observed crossing between 10^{-4} and 2×10^{-4} ; the same arithmetic at $d = 7$ (p^3 channel vs p^4 MWPM) predicts $p^* \approx 6 \times 10^{-5}$ — below the measured range, which is why CM(1) still beats MWPM at 4×10^{-4} there.

4. The microscopic mechanism: phantom chains

Anatomy census of the 359 pairs that defeat only CM(1) ($d=5$, $p=2 \times 10^{-4}$):

property	census
fault composition (detector counts)	(2,4): 113, (2,3): 72, (3,4): 66, (4,4): 48, (3,3): 34, rest: 26 — ≥ 1 correlated hyperedge in 347/359 (97%)
physical cause (all equivalent circuit explanations)	359/359 pairs: both mechanisms caused by DEPOLARIZE2 (CNOT noise); 0 pairs explainable without it. Data-qubit DEPOLARIZE1 Y's (vertical hyperedges, no space-time reach) and measurement/reset flips (single-sector, time-like) never assemble a failure.
geometry of the failure	45% of pairs overlap on ≥ 1 detector (ambiguous residual syndrome); CM's wrong matching differs from MWPM's by a median of 4 edges; 64% flip the observable through a boundary edge the honest matching does not use.

The template. (1) A CNOT-propagated correlated fault (Y or mixed two-qubit error; its decomposition is one X-edge *plus* one Z-edge with diagonal space-time reach) is selected — correctly — in the first pass. (2) Believing it with certainty discounts every hyperedge partner of every selected edge to $O(1)$ weight: a cloud of $\sim$ a dozen phantom edges, of which at most one corresponds to anything that fired. (3) A second fault leaves a detector whose honest route costs $\sim \log(1/p)$ per edge; the phantom cloud offers an alternative at $O(1)$ per edge that exits through a differently-flagged boundary edge $\rightarrow$ the logical prediction flips. Worked example (faults on edges (5,24) + boundary hyperedge (14,15,20)): MWPM selects the true explanation at base cost 23.0; CM(1) prefers a detour of base cost 30.3 — $e^{7.3} \approx 1450 \times$ less likely under the true model — because its *reweighted* cost is lower. The logical operator is thus crossed by 2 real faults + ~ 4 decoder-donated free edges: the phantom segment pays for the missing third fault. This is why d_{eff} drops by one unit.

5. The trade-off, mathematically

Expand the LER in fault order, $n = \lceil d/2 \rceil$ ($= 3$ at $d = 5$):

$$\text{LER}(\alpha, p) = [C - B(\alpha)] p^n + H(\alpha, p) p^{n-1}$$

Benefit sector $B(\alpha)$ (correlation correction): genuine correlated faults live in the same p^n sector as MWPM failures; CM lowers its coefficient by tipping near-degenerate matchings. B has an internal, p -independent optimum $A < 1$ — even at full strength some equal-order decisions get over-tipped. **Harm sector $H(\alpha, p)$** (distance loss): the phantom channel enters one fault order lower. A spurious detour of k boosted edges beats an honest margin $\Delta \approx a \cdot \log(1/p)$ iff

$$k(|\log \alpha| + c) < a \log(1/p) \Leftrightarrow \alpha > \alpha_{\text{th}} \propto p^{a/k},$$

so H is a *threshold staircase* whose steps slide toward $\alpha = 0$ as p decreases. Measured staircase (failing pairs vs α , exhaustive enumeration):

α	$p=10^{-4}$	$p=2 \times 10^{-4}$	$p=10^{-3}$	$p=4 \times 10^{-3}$
0.05	2	2	0	0
0.1	10	6	0	0
0.2	36	14	2	0
0.3	70	54	8	3

α	$p=10^{-4}$	$p=2 \times 10^{-4}$	$p=10^{-3}$	$p=4 \times 10^{-3}$
0.5	188	154	82	56
1.0	423	373	307	268

The onset α_{th} shifts by $\approx \sqrt{10}$ per decade of p , i.e. $a/k \approx \frac{1}{2}$. Note H depends on p only through the *weight landscape* ($\log(1/p)$ in the θ -functions), a slow logarithmic staircase — the occurrence probability of configurations is entirely in the p^{n-1} prefactor, so nothing is double-counted; at $\alpha = 1$ (all thresholds crossed) H is nearly constant ($423 \rightarrow 268$ while p^2 sweeps $\times 1600$). This p -dependence is essential: with fixed thresholds α^* would stall at a constant instead of decaying.

Minimising over α gives $B'(\alpha^*) = \partial_\alpha H/p$, hence two regimes: for $p \gg p_0$ the threshold cloud sits above A , $\partial_\alpha H \approx 0$, and $\alpha^* = A$ (plateau); for $p \ll p_0$ the $1/p$ -amplified staircase pins α^* to the sliding threshold cloud, $\alpha^* \propto p^{a/k} \approx \sqrt{p}$. The saturating law is the minimal interpolation of these asymptotics:

$$\alpha^*(p) = \frac{L}{1 + (p_0/p)^m}, \quad L = A \approx 0.75, \quad m = a/k \approx \frac{1}{2}, \quad p_0 \approx 7 \times 10^{-4}.$$

fit parameter	mechanistic identity	check
plateau $L = A \approx 0.75$	internal optimum of the benefit sector (p -, d -independent)	0.775 ($d=5$) vs 0.749 ($d=7$)
tail $m = a/k \approx \frac{1}{2}$	margin-to-boost-count geometry of the dominant detours	staircase shifts $\sqrt{10}$ /decade; swept α^* tail $\approx p^{0.6}$
crossover $p_0 \approx 7 \times 10^{-4}$	W_2 at $\alpha = A$ becomes comparable to the p^3 benefit stakes	where CM(1)/MWPM peels away from best- α /MWPM

6. Control experiments

(a) Code capacity (data depolarization only, $r = 2$): harm off, benefit on. The enumeration logic predicts an empty phantom channel (no CNOT-propagated reach). The α -sweep confirms every signature: CM(1) beats MWPM at every (d, p) ; α^* is interior (0.3–0.7) but p -independent — no collapse, no saturating law; the LER(α) valley is shallow (2–5%). The α -dependence that remains is pure benefit-sector shape: at $\alpha = 0.1$ and $p \geq 0.05$ the posterior $\alpha \cdot P(\mu|\nu)$ falls below the prior and the keep-the-prior rule makes CM *bit-identical* to MWPM (16630/16630 errors), while at $\alpha = 1$ equal-order over-tipping costs a few percent. This is the two-sector model with $H \equiv 0$.

(b) Shot-level event census ($d=5$, $p=4 \times 10^{-4}$, 2×10^7 shots, common random numbers). Errors MWPM/CM/Pearl = 154/114/88. CM-overcorrections (CM wrong where MWPM right): 51; CM-helps: 91 — correlations net-help, which is why $\alpha^* > 0$. Pearl(0.2) recovers Case-A overcorrections (41 shots, rate 2.1×10^{-6}) while keeping Case-B repairs (79, 4.0×10^{-6}); in 2 shots Pearl is right where *both* baselines fail. 3-D space-time visualisations of all three event types are in the notebooks.

(c) Real hardware (Google qec3v5, $d=5$, basis X, $r=5$; single-configuration probe). LER: MWPM 0.148, CM 0.128 (94.8% per-shot agreement with the released correlated-matching file), belief matching 0.113, tensor network 0.108. A validation/test α -split gives $\alpha^* = 0.85$ and Pearl test-LER 0.1258 vs CM 0.1265 — the device operates near threshold, squarely in the plateau regime where damping buys almost nothing, exactly as the law predicts. The damped-CM advantage lives at lower effective p than this hardware reaches.

7. Predictions and implications

- **d-scaling:** hard CM loses $\approx$ one unit of fault distance ($3 \rightarrow 2$ at $d=5$, $4 \rightarrow 3$ at $d=7$); the CM-below-MWPM crossover should appear at $p^* \approx 6 \times 10^{-5}$ for $d = 7$, and $d = 9$ should show $5 \rightarrow 4$ (testable by 3-fault enumeration or slope).
- **Noise-model knob:** the phantom channel is 100% DEPOLARIZE2-built. Suppressing two-qubit-gate noise (or biasing it away from Y/mixed components) should empty the channel: CM(1) then keeps full distance and $\alpha^*(p)$ stays at its plateau for all p — a single-knob falsification test.
- **Shot-free α calibration:** α^* costs 3×10^9 shots per point by sweep, but minutes by enumeration: $W_2(\alpha)$ exactly (all pairs), $W_3(\alpha)$ by importance-sampling triples $\alpha \ p; p; p_k$, then $\text{argmin}[W_2 + W_3]$. Accuracy improves as $p \rightarrow 0$, exactly where sweeps become impossible.
- **Adaptive decoding:** since α^* is p -dependent (and drift makes p time-dependent), no fixed strength is right — the mechanistic ground for per-shot adaptive reweighting (the GNN-SAC policy), whose learned discounts were independently observed to be weaker than CM's, i.e. damping discovered from reward alone.

Appendix: result-to-artifact map

result	artifact (repo path)
α sweeps (circuit level, $d=5,7$ + low- p $d=7$; code capacity)	data/alpha_sweep_extended_fixed_MWPM.csv, ..._d7_lowp.csv, ..._code_capacity.csv; scripts/test_alpha_decode.py
$\alpha^*(p)$ extraction, fits, penalized model comparison	notebooks/alpha_best_functional_relation.ipynb

result	artifact (repo path)
exhaustive 1-/2-fault enumeration; α -threshold staircase	scripts/enumerate_two_fault_failures.py; scripts/enumerate_alpha_threshold.py
worked 2-fault examples, ground truth + error types, 3-D	notebooks/cm_confusing_two_fault_examples.ipynb
shot-level Case A/B/C census + 3-D space-time views	notebooks/pearl_vs_cm_shot_level_3d.ipynb
Pearl $\equiv$ damped; Fowler $\equiv$ conditional	notebooks/pearl_vs_damped_grid.ipynb; notebooks/ fowler_vs_conditional_reweighting.ipynb; docs/fowler_vs_conditional_reweighting.md
saturating-law derivation (long form)	docs/saturating_law_mechanism.md / .pdf